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Rule capture of automated compliance checking of building requirements: a review

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In the architectural, engineering and construction industry, building design needs to be checked against regulations before it can be finalised and progress to the construction stage. The traditional manual compliance-checking process is error prone and time consuming. As a solution, automated compliance checking (ACC) was proposed. Rule capture is a crucial bottleneck of ACC. Despite many studies in this domain, no research has synthesised the themes and identified future research opportunities. This paper aims to fill this gap by conducting a systematic literature review and identifying challenges facing this field. The findings revealed that the rule capture process had attracted interest in the past years, and more semi-automated and automated methods have been proposed. The current representation development process lacks a methodological backdrop. The existing representations cannot represent ‘unknowns’ and ‘side effects’, lack the ability to deal with ambiguous rules and are typically restricted by the rule engine and/or target data model. The understanding of rules, representations and the relationships between them is insufficient. Further research is required to address these issues.

Keywords: Building Information Modelling (BIM)/construction management/design/UN SDG 9: Industry, innovation and infrastructure

1. Introduction

1.1 Background

In the architectural, engineering and construction (AEC) industry, ensuring design compliance is mandatory before progressing to the construction stage (Soliman-Junior *et al.*, 2021). However, the current manual compliance-checking process is often error prone, time consuming and costly (Han *et al.*, 1998; Macit İlal and Günaydın, 2017). Therefore, automated compliance checking (ACC) has been actively researched as a promising solution. It generally includes four steps: rule interpretation, target model preparation, rule checking and reporting (Eastman *et al.*, 2009).

There have been two approaches to achieving ACC: the target model and the rule capture approach. The target model approach primarily aims to find suitable methods for easy and efficient design data retrieval and query (Solihin *et al.*, 2020). While these studies typically develop new methods for easier retrieval and query data, they tend to pay less attention to the accuracy and expressiveness of rule representation (i.e. rule representations are sometimes restricted by the structure of the target data model). In contrast, rule capture studies aim to develop a computer-readable representation for building rules with minimum knowledge loss (Solihin and Eastman, 2016). These studies develop methods to interpret rules written in natural language to a suitable computer-readable form, typically requiring a deep understanding of rule

structure, semantics and complexity (Hjelseth and Nisbet, 2011). Due to the expertise required, rule capture is often conducted manually, making it a time-consuming and costly bottleneck in the ACC process (Solihin and Eastman, 2016). This step is also crucial, as any inaccurate rule capture will likely result in non-compliance.

The existing literature review mainly focuses on ACC system development (Eastman *et al.*, 2009) or the target data model essential for ACC (Hu *et al.*, 2021). Despite significant research efforts, research on the rule capture for ACC has yet to be synthesised. As the rule interpretation process is still a time-consuming bottleneck that impedes the efficiency and accuracy of ACC, it is essential to understand the current progress and existing issues to help the research community work towards problem solving. This research aims to review systematically the rule capture literature to find current research themes and existing challenges. Specifically, the research questions of this study are as follows: (a) what are the themes of ACC research in the rule capture process? and (b) what are the challenges and future research directions of ACC from the rule capture perspective?

1.2 Related work

In the past decades, numerous studies have been conducted on various aspects related to ACC, including rule capture

(Yurchyshyna and Zarli, 2009), preparing design data for ACC (Solihin *et al.*, 2017; Wu and Zhang, 2022) and developing rule-checking software (Martins and Monteiro, 2013).

Some research reviewed these ACC studies. For example, Eastman *et al.* (2009) evaluated the functional capabilities of five major ACC systems available at the time and identified major problems of ACC system development. Amor and Dimyadi (2021) reviewed some recent ACC developments, but they did not prioritise the rule capture perspective and only touched on some of the rule representation methods. Noardo *et al.* (2022) reviewed literature related to the digital building permit process. ACC studies related to building performance and planning have been reviewed. However, a detailed analysis of the rule capture perspective of ACC is lacking. Noardo *et al.* (2022) also highlighted the lack of literature review, evaluation of software or procedure and data review research in this domain. Consistent with this finding, despite rule capture being a time-consuming bottleneck of ACC (Solihin and Eastman, 2016), no research has reviewed studies on this topic to understand the current progress and propose future directions. This research thus aims to address this research gap by providing a literature review on rule capture following a more rigorous process.

The remainder of this paper is organised as follows: Section 2 introduces the method of this paper. Sections 3 and 4 provide the quantitative and qualitative analysis results, respectively. Section 5 discusses the findings. Section 6 concludes this paper.

2. Methodology

This research adopted a systematic literature review method. It is widely adopted to synthesise and evaluate the state of knowledge, identify research gaps and create research agendas in a research domain (Snyder, 2019). It has been used in multiple previous studies in the AEC domain (Yin *et al.*, 2019). The whole process is shown in Figure 1.

The literature search was conducted using the Scopus database. Scopus was selected because it had the largest amount of literature related to this topic compared with other commonly used literature databases such as the Web of Science. It was also widely adopted by researchers in the AEC industry (Jin *et al.*, 2018). The authors ran a query in Scopus to retrieve peer-reviewed English-language journal articles and conference papers related to ACC. No year limit was applied to the data retrieval. First, the authors used frequently mentioned keywords regarding ACC to run the search in titles, abstracts and keywords of the literature. The keywords included

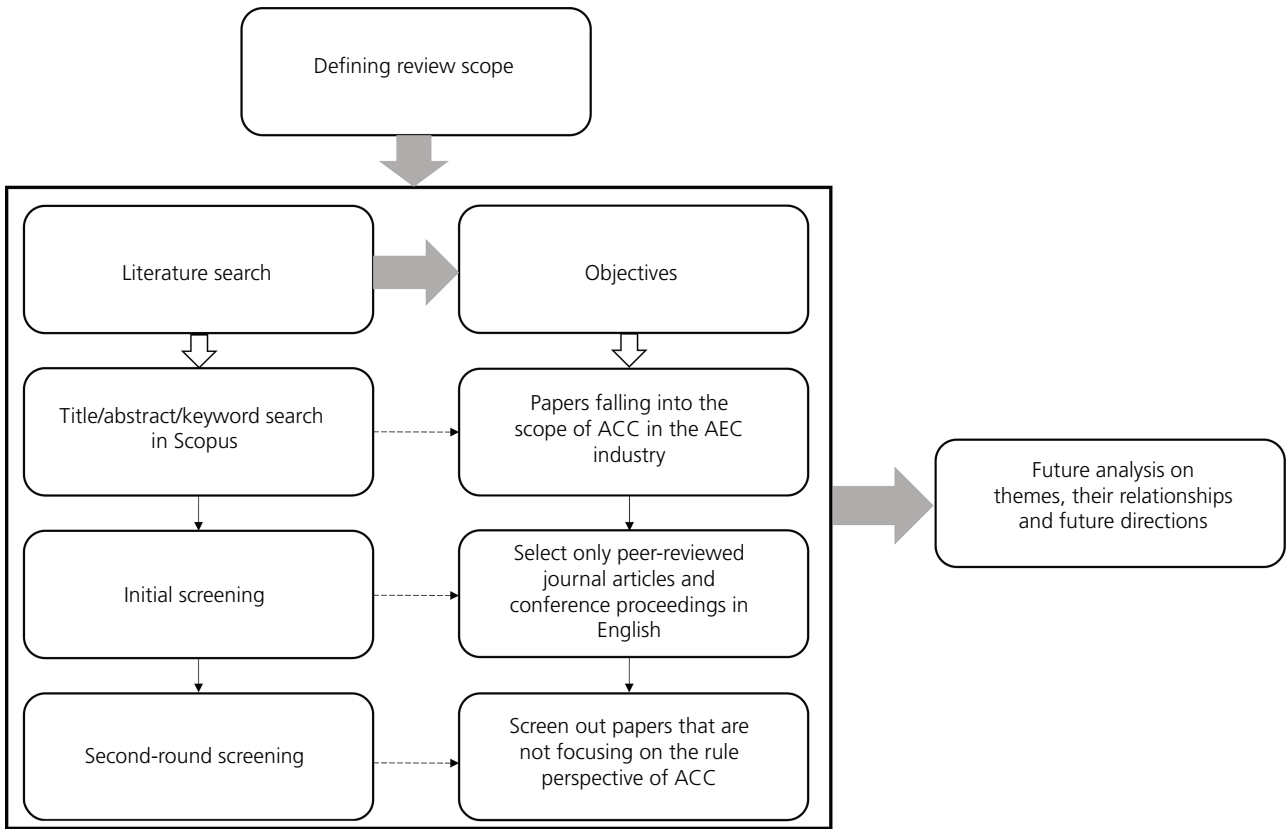


Figure 1. Literature retrieval and analysis process

‘automatic compliance checking’, ‘automated compliance checking’, ‘code compliance checking’, ‘rule-based design check’ and ‘regulation compliance check*’, where ‘*’ was a wildcard character that could mean any letters. These keywords were run with the search query "construction" OR "architecture" OR "building" to limit the scope to the AEC industry. Then, peer-reviewed journal articles and conference proceedings written in English were selected. As of 10 January 2023, 640 papers were found.

Two rounds of screening were conducted to refine the papers retrieved. Two types of papers were excluded after a scan of their titles, abstracts and keywords: (a) those not relevant to the AEC industry and (b) those not focusing on the rule capture perspective of ACC (e.g. only focusing on design data integration and model preparation or focusing on the adoption of ACC). Thus, the number of relevant papers was reduced to 124.

The full texts of the shortlisted papers were then reviewed and analysed in detail. Specifically, the authors first analysed the papers quantitatively by looking at the number of publications by year and by manual or automated rule capture per year. Then, the paper contents were analysed in depth to identify current trends and future research directions.

3. Quantitative analysis results

3.1 Number of publications by year

Figure 2 shows the number of rule capture publications each year. There have been papers published on the rule capture topic since 1996. However, for 20 years, this topic gained little popularity, and only in 2015 did researchers start being interested in this topic. Since 2015, each year, there have been nine or more papers published on this topic, whereas in 2022, the number of publications reached the highest at 18. This recent popularity gain could result from the increasing adoption of building information modelling (BIM), which facilitates the ACC process.

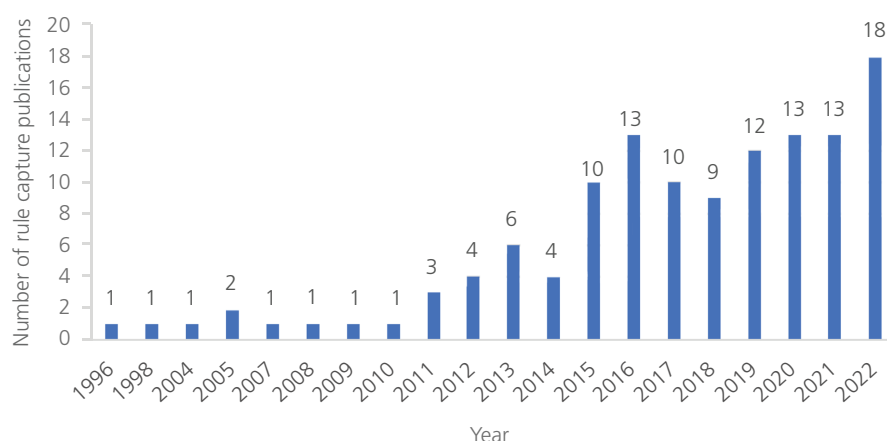


Figure 2. Number of rule capture publications by year

3.2 Yearly publications by manual or semi-automated and automated rule capture process

As shown in Figure 3, most rule capture studies (77 out of 124) are based on a manual method. Since 2011, some researchers have focused on using semi-automated or automated methods for rule capture. In 2014, the number of publications on rule capture using semi-automated or automated methods overtook the number of those on manual methods for the first time. The highest number of yearly publications on semi-automated and automated rule capture was 10, achieved in 2021.

4. Qualitative analysis results

4.1 Manual rule capture process

4.1.1 Rule classification

To assist the rule capture process, scholars have developed many classification methods. Many studies explored the rule semantic structure. Macit İlal and Günaydın (2017) considered rule dependency, where rules were classified into self-contained and linked explanatory categories. The former refers to rules that are complete in themselves and do not depend on other rules, while the latter links to other rules as clarifications, exceptions, exemptions or modifications. Nawari (2020) provided a more detailed classification with conditional, content, ambiguous and dependent clauses. Conditional clauses can be directly interpreted into computable rules. Content clauses are definitions or clarifications that do not translate into decisions. Ambiguous clauses use subjective wordings such as ‘big enough’ or ‘relatively’. Moreover, dependent clauses mean whether the clauses need to be checked depending on the outcome of other clauses. Dependent clauses are suggested to be checked manually due to dependency. This classification method recognised the ambiguity and dependency of building rules.

Soliman-Junior *et al.* (2020) classified rules against the nature of requirement (i.e. qualitative, quantitative and whether the

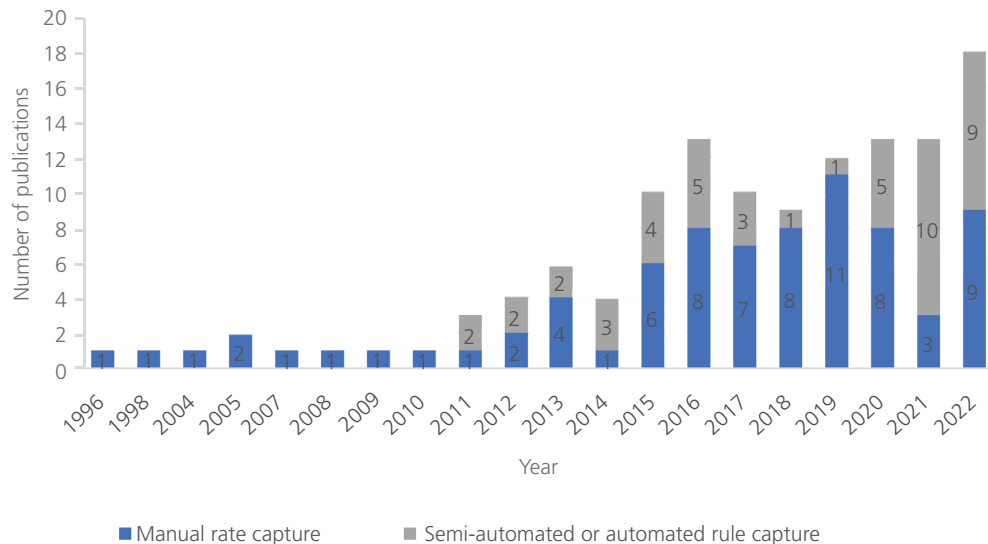


Figure 3. Yearly publications by manual or semi-automated and automated rule capture process

requirement is ambiguous) and the possibility of translation into logical requirement (i.e. logical, non-logical). This method attempts to explore the relationships between the type of rule and whether it can be translated into a machine-readable form but fails to reveal the characteristics that make the rules possible or impossible to be interpreted into logical rules. ‘Non-logical’ rules are suggested to be checked manually.

Yurchyshyna and Zarli (2009) introduced a classification method based on interpretability related to the industry foundation classes (IFC) model. The four types of requirements include (a) completely IFC interpretable; (b) reformulated in IFC (requires human interpretation); (c) partially interpretable (the IFC model is not sufficient for interpretation); and (d) non-interpretable (rules with ambiguous information or describing common knowledge). A major deficiency of this approach is that it restrained the rule representation to the IFC model, which is often insufficient to represent all required information in the rules, particularly abstract geometrical and topological constraints (Preidel and Borrmann, 2016). As a result, too many rules are left to be checked manually.

The above classification methods are typically used to distinguish which rules are suitable for ACC and which are suitable for manual checking. For this reason, these studies tend to have a selective bias: only rule features that are not conducive to ACC become the main focus, such as dependency and conditionality, but the general features of rule provisions (e.g. semantic constructs and logic connectives) are unrevealed.

Some studies have attempted to explore general rule features. Hjelseth and Nisbet (2011) identified four general constructs in each rule clause: requirement, applicability, selection and

exception (RASE). They also recognised the influence of logical connectives among different semantic constructs on the checking results and used a tree-like method to demonstrate the logical calculus (Nisbet *et al.*, 2009). In addition, the RASE method developed a dictionary to (a) link the terms in the regulation representation and in the target model; (b) maintain the consistency of terms; and (c) deal with algorithmic calculations and simulations. The later extension of RASE further recognised the need to capture the actions when the rules have outcomes other than pass/fail (Beach *et al.*, 2015). It incorporated ‘output’ and ‘total’ constructs to represent the ‘actions’ in Building Research Establishment Environmental Assessment Method (Breeam) rules. Solihin and Eastman (2016) also generalised rule structure logic, including concepts, requirements, constraints, functions, derived data, exceptions and rule dependency. Notably, functions and derived data are implicit in rule provisions. Rule analysts capture them when turning the ‘black boxes’ into operable ‘white boxes’. This highlights the importance of going beyond the syntax, language and grammar of rules to understand the complexity of the rules.

Considering the rule complexity about data structure, Solihin and Eastman (2015) developed a four-class classification method: (a) class 1, rules that require a single or a small number of detailed data items; (b) class 2, rules that require simple derived attribute values; (c) class 3, rules that require extended data structure; and (d) class 4, rules that require a ‘proof of solution’. This classification took a holistic view of the ACC system, where the rule representation is later mapped to a model view definition of the target model. This method provides insights into the objects, properties and relationships that need to be checked and the geometrical, mathematical algorithms and simulations required when executing rules. However, such a representation again

suffers from the limited expressiveness of IFC entities. A recent study proposed a four-criteria classification categorising automatically checkable building rules based on their semantic constructs and intensity and whether they are self-contained or linked explanatory and prescriptive or performance based (Zhang *et al.*, 2022a). However, they claimed that quality and aesthetic rules could only be manually checked, which needs to be verified further.

Summarising the existing literature, despite the many existing classifications, most are categorised using a single dimension or a few casually selected criteria or metrics. A consolidated list of metrics could better classify rules, thereby facilitating the rule capture process.

4.1.2 Rule representation

The first category of rule representations is the production rule. For example, Balaban *et al.* (2012) developed an ACC system for Turkish fire codes. They used pseudo-codes in the form of if-then to represent the rules and implement the rule checking in Java language. The stand-alone and hard-coded nature of this system made it difficult to be expanded to other disciplines and establish a submission workflow for checking the design model. Tan *et al.* (2007, 2010) proposed a decision table approach. Parameters (e.g. location, type) extracted from rules are used as subheadings for the decision table. The reference index is used to denote the original link or cross-reference. However, this decision table can deal only with rules with similar conditions and actions. In addition, it still fails to show the logical relationships among rules.

Another category of representation is a logic-based method. Parametric tables in some commercial software program, such as the Solibri Model Checker software, can be found in this category. However, they are not technically a separate representation due to rules embedded in the rule engine. This method often lacks transparency and is difficult to maintain (Solihin and Eastman, 2016). Some studies have adopted predicate logic to represent building rules (Kerrigan and Law, 2005). However, predicate logic can become lengthy and hard to read when dealing with complex rules. Solihin and Eastman (2016) extended a conceptual graph approach to represent building rules to address this issue. It has a foundation in predicate logic but is tailored to the requirements of building rules using IFC properties and objects. Its graphic notations improve readability. However, as it is based on predicate logic, it cannot represent actions other than true/false. In addition, the use of IFC in representing rules restrains this method to the target data model. As a result, only a limited number of rules can be represented using this approach.

Representations based on the semantic web have gained popularity in recent years. One of the earliest attempts using the semantic web was that by Pauwels *et al.* (2011), where N3 logic was used to represent building rules. It can express both the data 'for all' or 'for some', such as the quantifiers in predicate logic.

Some studies have adopted Semantic Web Rule Language (SWRL). It can help infer new knowledge from the Web Ontology Language (OWL) ontology. For example, Hu *et al.* (2013) used OWL and SWRL axioms to represent constraint knowledge on construction quality. Beach *et al.* (2015) interpreted building rules to SWRL assisted by RASE. Their mapping ontology helps in the alignment between data and regulation ontologies, ensuring that each ontology is independent of one other.

Some more recent studies used Sparql Protocol and RDF Query Language (Sparql). In the paper by Zhong *et al.* (2018), design constraints were represented as Sparql queries, which were used to query logic facts constructed from multiple data sources to monitor building environments. Similarly, Zhang *et al.* (2018) developed BimSparql based on Sparql and Sparql Inference Notation (Spin) to represent building requirements related to geometry. Also using Spin and Sparql, Xu and Cai (2020) achieved ACC of spatial relationships in underground utilities. Another research by Jiang *et al.* (2022a) used Jena, an open-source Java application programming interface for the semantic web, to develop a rule representation method independent of the data model. Jena provides rule inference capabilities. Using Jena, customised rules can be implemented on the Resource Description Framework (RDF) graph and generate new triples. While writing Jena rules is straightforward, Jena rules can only represent logical conjunctions (i.e. and), which makes it unsuitable for rules with logical disjunctions (i.e. or) or negations (i.e. no). The proposed representation was also unable to address geometric rules.

Among these representations, Sparql is the most semantically expressive, and its Construct queries have the expressive power of first-order logic (Kostylev *et al.*, 2015). It also has predefined aggregate algebra functions to help with rules that require simple calculations, such as the counts of fire exit doors. However, there are currently two obstacles to adopting semantic web technology for ACC. Firstly, the representation of geometric rules is still difficult. Although Zhang *et al.* (2018) demonstrated the use of Sparql extensions to represent geometric rules, it only covers limited geometric capabilities, and some geometry (e.g. curves) cannot be queried. Secondly, the learning curve for the semantic web is quite steep, which makes it hard for non-programmers to learn.

The last category is language-based methods. Focusing on performance-based rules, Dimyadi *et al.* (2014) encoded compliant design procedures based on Business Process Model and Notation (BPMN). They used the Drools rule language to represent building rules. Similarly, Häußler *et al.* (2021) used BPMN and Decision Model and Notation (DMN) to represent German railway codes. Of the examined rules, 46% were found to be implementable using BPMN and DMN. Wang *et al.* (2022) represented a mechanical, electrical and plumbing (MEP) design model and the logical relationships in MEP requirements using Neo4j graphs. They performed ACC using the Cypher query language to compare the two graphs.

With the recognition that general-purpose languages are not expressive enough to represent the domain knowledge in the AEC industry, several domain-specific languages have been proposed, including text-based languages and visual programming languages (VPLs). The Building Environment Rule and Analysis (Bera) language (Lee *et al.*, 2014) for building circulation and spatial rules falls into the former group. It is a domain-specific query language where the syntax and functions embedded in building rules can be represented. Therefore, it is easier to learn for non-programmers compared with general-purpose languages. However, it was mainly developed for checking spatial rules; thus, the syntax is not sufficient for other types of building rules. BIM Rule Language (BIMRL) is another domain-specific query language proposed by Solihin *et al.* (2020). Its development was greatly influenced by Structured Query Language (SQL). Querying using BIMRL is easier than querying using SQL, as a simplified schema was applied to improve query performance. BIMRL provides good expressiveness and can address geometrically complex rules, where temporary geometries are used to assist geometry and spatial checking. However, its weakness lies in its requirements for knowledge of query languages. Other studies that used a similar approach are those by Sydora and Stroulia (2019, 2020), where a domain-specific rule language for interior design layout rules was proposed. This language incorporates reporting the severity of rule failure, which is helpful for the assessment and comparison in a generative design use case. Nevertheless, like Bera by Lee *et al.* (2014), it also suffered from the limited expressiveness issue, as it was developed mainly for interior design rules and was only equipped with limited capability for ACC. There is also no evidence of its extensibility and scalability for other rules.

Other research studies used VPL (Kim *et al.*, 2019; Preidel and Borrmann, 2016, 2017) to represent rules. Either purpose-built software or extensions of existing visual programming software (e.g. Dynamo) were proposed. These languages can represent complex rule logic without intensive computer programming (Solihin and Eastman, 2016). For example, Kincelova *et al.* (2020) and Villaschi *et al.* (2022) used Revit Dynamo scripts to represent building codes. However, only design models in the proprietary '.rvt' format can be checked. Based on open BIM, the Visual Code Checking Language was developed, where small 'white boxes' with known functions were linked to input and output ports by wires, making the rule-checking process transparent and easy to understand (Preidel and Borrmann, 2015). KBim Visual Language (Kim *et al.*, 2017) is a similar effort using VPL to represent Korean building act regulations. It was developed based on a scripting language called KBimCode (Lee *et al.*, 2016; Park and Lee, 2016a, 2016b; Park *et al.*, 2016). An advantage was that the representation was independent of the design model. Similarly, Zhou *et al.* (2018) represented envelope energy efficiency codes based on their VPL. Notwithstanding, due to their nature, most VPLs seem to have deficiencies when handling recursions in rules. In addition, each of the existing VPLs developed functions only for their own country's design codes.

Nevertheless, the expressiveness of current representations is not sufficient to represent various types of rules fully. Moreover, many representations are difficult for domain experts to understand.

4.1.3 Rule organisation

The aforementioned rule representation methods mainly focus on single rule provisions, while few studies have explicitly considered rule organisation (i.e. the relationships and hierarchies among rule provisions). However, it is essential to consider and represent rule dependencies because rules can be better organised, but the relationships and interdependencies affect the checking results.

One of the earliest rule organisation methods is the standards analysis, synthesis and evaluation (SASE) model (Fenves *et al.*, 1987). It includes two networks on four levels. The network on the top level is the organisational network (organising building codes). The information network (dependency relationships among provisions) includes an individual provision level (decision tables), derived data item level and basic data item level. However, this method does not apply to data items and has complex precedence relationships (Macit İlal and Günaydin, 2017).

There have been some attempts based on object-oriented thinking. For example, Gowri and Depanni (1998) organised rules based on objects and their required attributes. Building codes were represented using simple production rules with a dependency network to denote their interrelationships. However, the rule representation was dependent on the target model. Yang and Xu (2004) also employed an object-oriented modelling method to organise rules, but their ACC system was designed to work with computer-aided design models. More recently, Malsane *et al.* (2015) explored parts B1 and B2 of England's building regulations based on object-oriented thinking and recognised that entities in the target IFC schema were insufficient to represent the requirements. However, they only outlined the development process, and no representation was developed.

Yurchyshyna *et al.* (2008) and Yurchyshyna and Zarli (2009) recognised the issues of traditional regulation text organisation. Instead of organising by themes, they suggested that rules be organised based on their contents. They developed three main classifications to organise rules – namely, classification (a) by construction, (b) by key concepts and (c) by application conditions. However, this approach failed to address the interdependencies among rule provisions.

Integrating SASE with RASE, Macit İlal and Günaydin (2017) proposed a method of organising rules on four levels: domain level, rule level, rule-set level and management level. The domain level is a library of concepts, attributes and definitions to avoid repeating definitions when representing rules. The rule-set level stresses the logical connectives among objects that appear in rules. Finally, the management level recognises the importance of

the overall organisation of building rules; it does so by grouping closely related rule sets (i.e. those with the same concepts) and connecting them. Nevertheless, this model is programming intensive and requires the involvement of a software engineer.

Recently, Jiang *et al.* (2022b) proposed a representation including concept, clause and code levels. The concept level provides concept definitions. The clause level represents the semantic meaning and logical relationships within a clause. Clauses are organised by the cross-references and semantic correlations on the code level. However, the rules are represented using the Jena rule syntax and Sparql queries, which have limitations in representing geometric rules.

4.2 Semi-automated or automated rule capture process

An emerging trend in the ACC domain is using machine learning (ML) (particularly natural language processing (NLP) techniques) to help with various tasks in the rule interpretation and representation process, such as text classification (TC), information extraction (IE) and semantic alignment. Sections 4.2.1–4.2.5 present a review of the related work.

4.2.1 Semi-automated or automated rule classification

To assist automated rule interpretation and provide enhanced IE, some studies have developed automated TC methods to classify building rules. For example, Salama and El-Gohary (2011) compared supervised and unsupervised ML methods to classify rule clauses based on their content. They later proposed using supervised ML and a deontic model to perform scope-based (e.g. risk management) TC, which achieved 96% precision and 100% recall (Salama and El-Gohary, 2016). Also, based on the semantic meaning of clauses, Zhou and El-Gohary (2016) proposed a domain-specific TC method, where they developed a four-hierarchy classification of environmental building codes and used ML algorithms to conduct the automated TC. The model yielded a recall and a precision of 97 and 84%, respectively. However, this study focused only on environmentally related rules and may not classify rules in other disciplines.

While the aforementioned research classifies requirements based on the contents, Zhang and El-Gohary (2018, 2021a) considered a more comprehensive set of metrics, including syntactic metrics, semantic features and computability metrics. Twelve types were identified using a clustering-based approach based on these metrics. However, this work only analysed the International Building Code, while analysis of other regulation documents could find more syntactic, semantic and computability metrics. Furthermore, these metrics were tested only in the IE task but not in other tasks, such as rule representation.

4.2.2 Semi-automated or automated IE

Accurate IE is a crucial step to rule representation. There have been studies to finish this step automatically. Some studies used deep-learning (DL)-based NLP methods. For example, Song *et al.* (2018) used NLP techniques to extract nouns from the Korean

Building Act and adopted DL to analyse their semantics. The nouns were then converted to objects and properties as in the design model. However, this method only extracted nouns in the regulations. In addition, it limited the expressiveness of the representation of regulations, as the regulations can only exist in the same form (i.e. using the same set of objects and properties) as the data model. Their later work developed a method using a bidirectional long short-term memory model with conditional random fields to extract the predicate–argument structure of building regulations (Song *et al.*, 2020). However, this method achieved a validation accuracy of only 77.4% with 50 epochs. Another research proposed a deep IE method using deep neural networks, where the syntactic and semantic information elements were extracted from construction regulatory documents (Zhang and El-Gohary, 2021b). This method benefited from transfer learning, which reduced the time to prepare the data set. Similarly, Li *et al.* (2020) also dealt with a complex extraction problem (i.e. overlapping relations for the same entity) by proposing a joint extraction method, where a hybrid DL algorithm and a decomposition strategy were used. The proposed method yielded a precision, a recall and an *F*-measure of 88.08, 85.19 and 86.61%, respectively.

Other studies used semantic-based methods. For example, Zhang and El-Gohary (2016a) developed a semantic rule-based NLP method for automatic IE from building codes. This proposed method used a series of pattern-matching and conflict resolution IE rules, an ontology of concepts and relations and achieved a precision of 96.9% and a recall of 94.4%. Despite the promising result, this IE method was tested only on quantitative requirements.

More recently, Zhou and El-Gohary (2017) proposed an ontology-based algorithm for IE of requirements from energy conservation codes, where TC, domain-specific preprocessing and extraction techniques based on ontological pattern matching, dependency and cascade were employed. This method achieved 98.5% precision and 97.4% recall. Later, they presented an automatic method for complex extraction tasks (e.g. incomplete sentence structure and different levels of development (LODs)). Three sub-methods were developed, including domain-specific text splitting and stitching, incompleteness-aware dependency extraction and LOD detection methods (Zhou and El-Gohary, 2022). This method yielded 96.8% recall and 97.6% precision. Also using an ontology-based method, Zhong *et al.* (2022) performed IE of construction quality inspection regulations. They first identified and represented knowledge patterns using an ontology and then encoded the regulations into Java Annotation Patterns Engine rules. The experiment results on the prototype were satisfactory.

4.2.3 Semi-automated or automated rule organisation

Some studies have looked at the intrarelations and interrelations of clauses. Focusing on the hierarchical relationships of different components in a clause, Zhang and El-Gohary (2019, 2022a) developed an ML-based method for

extracting requirement dependencies and hierarchies within requirement clauses automatically. However, this method does not present the interrelationships among rule provisions.

Using Python packages, Zhou *et al.* (2021) addressed this issue by automatically transferring building codes from plain text to Extensible Markup Language (XML) and then to Neo4j graphs. The intermediate XML structure model helps present the cross-references among clauses, subsections and documents in Neo4j graphs. Using a rule-based relation extraction method, Xue *et al.* (2022a) developed the Building Code Network Generator (BCNG) to show an interactive visualisation of building rules. BCNG can present the hierarchy and cross-references among requirement clauses. It achieved a 99.4% precision, a 96.3% recall and a 97.8% F_1 score. However, rule-based methods typically lack scalability (Zhang and El-Gohary, 2019).

4.2.4 Semi-automated or automated semantic alignment

Some recent studies have realised the importance of keeping rule representation independent of the rule engine and target data model. This requires a corresponding semantic alignment method between the target design data and the rule representation. Some studies (e.g. Beach *et al.*, 2015) have proposed mapping ontologies to align them before the rule checking, while others have left the alignment to users when they conducted the ACC (e.g. Nawari, 2012). However, both methods rely on significant manual efforts.

In recent years, there have been some attempts to automate the alignment process. For example, Zhang and El-Gohary (2021c) proposed a semantic integration method for integrating concepts in BIM and requirements. The method is based on transformer-based language models (Bidirectional Encoder Representations from Transformers and Generative Pre-trained Transformer 2) and is fine-tuned using domain-specific corpora. Zhou and El-Gohary (2021) developed a two-level alignment method between the representations of energy regulations and the design model. The first level used domain knowledge and DL to match single regulation concepts to design model instances. The second level matched instance groups with requirement sentences using searching algorithms and network modelling. This method yielded a 93.4% recall and a 94.7% precision. More recently, Zheng *et al.* (2022) proposed a method for filling the semantic gaps between rule representations (in Sparql queries) and the design model. They established a domain knowledge ontology. This and unsupervised learning provided the knowledge base for semantic alignment and conflict resolution. This method achieved an accuracy score of 90.1%, outperforming the keyword-matching approach.

4.2.5 Semi-automated or automated rule representation

With the help of automated regulation TC, IE, rule organisation and semantic alignment, the recent years have seen many semi-automated or automated approaches to rule representation.

Some representations are logic based. For example, Salama and El-Gohary (2013) proposed a deontic model for representing building rules, where the rule provisions are written in deontic axioms. The model also shows the hierarchy of concepts and relationships among concepts in the requirements. Based on first-order logic, Zhang and El-Gohary (2016b) proposed a method for representing the design model as logic facts, which was checked against the logic rules (building regulations). However, only quantitative rules can be dealt with using the proposed method. Their later work developed a tree-shaped visualisation method to show the structure of a requirement (Zhang, 2017). While these representation methods focused only on text-based requirements, Xue *et al.* (2022b) proposed a logical representation for tabular rules. They developed an algorithm for detecting the layout of the tables and extract the rules semi-automatically. The rules were then transformed into logic rule representations automatically; 91.67% of tabular rules were processed correctly using this method.

Some studies have used semantic web technology to represent building codes semi-automatically or automatically. For example, Li *et al.* (2021) proposed a semi-automated method to interpret railway codes into an ontology. They extended ifcOWL to include railway code attribute values, properties and entities and converted the ifcOWL extension to the railway code ontology. Guo *et al.* (2021) automatically generated Sparql queries to represent building codes. Rules were extracted using NLP techniques and mapped to concepts or properties as in BIM models. The deficiency of this method is that the rule representation is not independent of the design model.

Zhang and El-Gohary (2022b) proposed an intelligent code to represent building codes using DL, which bypassed the IE and transformation processes that are often prone to errors. The proposed intelligent code is in the form of semantic requirement hierarchies and is readily usable by computers. However, this method seems to neglect interrelationships between rule clauses.

Nevertheless, despite the various methods developed for automated rule classification, IE, rule organisation, semantic alignment and rule representation, none can achieve 100% accuracy, which means that manual efforts are still required to review the results. In addition, most of them are based on NLP algorithms, which cannot handle ambiguous building requirements accurately (Doukari *et al.*, 2022).

5. Discussion

Many methods for rule capture have been proposed but with limited practical implementation. Although some commercial ACC solutions such as Solibri and Verifi3D can help with the ACC process (e.g. clash detection and check against geometric and property rules), they operate on proprietary rule engines and limited sets of available rules. As mentioned earlier, although there are some semi-automated and automated attempts, the rule

capture process either is not fully automated or requires experts' review, as 100% accuracy could not be achieved.

Transforming regulation texts to computer-readable rule representation (i.e. the rule capture process) can be divided into two steps, rule interpretation and rule representation. The rule interpretation step requires extensive experience and expertise in building rules. For this reason, it is primarily a manual process where domain experts analyse rules and take notes about the constructs, logic and implicit assumptions of rules. This time-consuming step can take up to 30% of the total time of implementing a rule (Solihin and Eastman, 2016). The importance of automating this step has been recognised, while attempts using ML methods (e.g. NLP) have not yielded very satisfactory results due to the complexity and diversity of regulation texts.

In contrast, the rule representation step may be easier to automate given that all the rule semantic constructs and logical connectives are already known. The RASE method (Nisbet *et al.*, 2009) is an example of automating this step. It provides a range of different representations (e.g. IFC constraints (Nisbet *et al.*, 2009), plain language (Hjelseth and Nisbet, 2011), Java Drools (Beach *et al.*, 2015), conceptual graph and visual programming (Nisbet *et al.*, 2022)), all generated automatically from the RASE mark-upped regulation text. However, this method fails to connect different types of rules and different representations. This is important, as not all representations are suitable for representing a specific type of building regulations. Although the study by Zhang *et al.* (2022a) attempted to match different types of building regulations with their suitable representations, a prototype needs to be developed to validate the feasibility of this approach.

The authors found that challenges (Figure 4) still exist in the rule capture process of ACC – namely, four challenges about current rule representations, one about the understanding of regulations

and one regarding the relationship between them – which necessitates further research. They are explained in Sections 5.1–5.6.

5.1 The rule representation development lacks a solid methodological backdrop

The design of a better rule representation method falls into the scope of design science research (Hevner and Chatterjee, 2010), where the rule representation method is the primary artefact being developed. Artefacts are typically developed based on specific objectives (Peppers *et al.*, 2006). The artefact development process requires assessment and evaluation of design options (i.e. different representation methods) before final design decisions are made (Pries-Heje *et al.*, 2008). Typically, the process also includes a few iterations.

However, in the current literature, frequently, new representations are arbitrarily borrowed from the available knowledge representation methods in other domains without a thorough analysis of the objectives and required functionalities for representation. As a result, none of the representation methods can represent all types of rules (Macit İlal and Günaydın, 2017). Furthermore, little research has adopted evaluation methods to assess design options before building the artefact. In addition, the current ex post evaluation of the artefact is mainly based on prototype development and validation by researchers using the prototype without input and feedback from actual users. While the demonstrations showed the usability of the artefact, the performance tests (e.g. efficiency) were often insufficient.

The artefact should be developed with more rigorous building and evaluation processes to address these issues. It is recommended to have both ex ante evaluations for design options and ex post evaluations to test artefact performance (Pries-Heje *et al.*, 2008). In this way, the usability, efficiency and accuracy of the proposed artefact could be better assessed. The proposed artefact has a

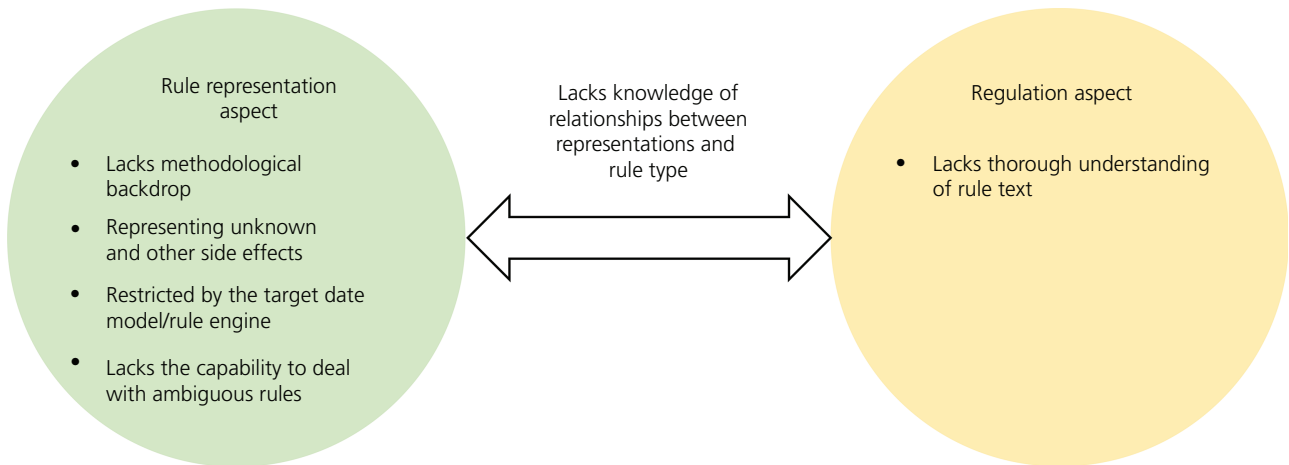


Figure 4. Challenges in the rule capture process of ACC

better chance of success before putting many resources into its development.

5.2 Reporting the ‘unknown’ and representing other side effects

Many of the existing representation methods (e.g. predicate logic, conceptual graph (Solihin and Eastman, 2016)) for building rules are drawn from the knowledge representation domain, a sub-domain of artificial intelligence. These methods are typically not tailored for building regulations and mostly deal only with binary results (i.e. true or false). However, in ACC of building regulations, as data are not always available and are not always needed in the model submitted to the ACC system, it is crucial to have ‘unknowns’ when the corresponding data cannot be found in the model provided (Zhang *et al.*, 2023a).

The binary results are underpinned by a closed-world assumption, which regards what is not known as false (Hustadt, 1994). Considering that not all required information is available in the building model, this can easily result in many ‘false negatives’, which compromises the accuracy and reliability of ACC.

Two methods have been considered to alleviate this issue. The first one is to ask the modeller to follow a modelling guideline for the convenience of checking. However, modellers typically find this a huge burden (Amor and Dimyadi, 2021). It would also affect the efficiency of model preparation and, eventually, the whole ACC process. In addition, much of the extra work incurred from this process may not be useful during checking later on.

The second method makes an open-world assumption. During the checking, the required information may not be available. Using an open-world assumption, when this happens, instead of an immediate ‘false’, it would be regarded as an unknown. Further questions may be asked, and the decision will be made based on the supplemented information. The second method saves resources, time and cost, as it requires information only when needed. However, currently, only few ACC rule engines support unknown values.

In addition, some rule clauses have side effects. For example, in Breeam, the final score is calculated by adding credits from satisfying individual clauses (BRE, 2018). Current methods drawn from other domains are not equipped with this function. For this reason, a domain-specific representation tailored for building regulations is needed. There have been attempts to adapt existing knowledge representation methods for building regulations. For example, Solihin and Eastman (2016) adapted the conceptual graph (Sowa, 1976) and proposed a new conceptual graph with features in line with BIM data. However, care must be taken when making such an adaptation. It could easily result in a lack of rigour because it may not be valid mathematically and logically.

5.3 Representations are restricted by the target model structure and rule engine

For easier information alignment, many existing representations are developed using the properties, objects and relationships from the target data model (e.g. IFC). However, the expressiveness of IFC is still limited. Some rules cannot be represented using the properties, objects and relationships from IFC, or it may be cumbersome to use IFC to represent them (e.g. circulation rules). This may limit the usability and efficiency of the ACC system.

In addition, some rule representations also suffer from restriction by the rule engine. For example, to adapt to a specific rule engine, the objects to be checked must be translated from natural language to the form acceptable in the rule engine to enable the alignment and checking process. Examples of such methods include query-based (Solihin *et al.*, 2020) and domain-specific-language-based (e.g. Lee *et al.*, 2014) representations with specific syntax and grammar customised to the specific rule engine.

The expressiveness issue calls for an intermediate way to link different ontologies, thus achieving the connection of the target data model and the rule representation. A viable way for this seems to be a dictionary-like approach (e.g. Nisbet *et al.*, 2009) using a customised and expandable dictionary (or ontology) to link the objects, properties and relationships in the data model (typically BIM model) to the rule representation. Such a method frees the representation from being restricted by the data model and rule engine structure. Although there are some existing mapping ontologies and semantic alignment methods, more research is needed to improve the alignment performance.

5.4 Representations lack the capability to deal with ambiguous rules

Researchers have pointed out that ambiguity in rules is a crucial hindrance to rule capture (Zhang and El-Gohary, 2022b). Ambiguous rules can take up to 53% of all building requirements (Soliman-Junior *et al.*, 2021). However, despite some attempts to understand and address ambiguity rules in ACC (e.g. Nawari, 2020; Zhang *et al.*, 2023b), no satisfactory results have been reported. The challenges mainly lie in the multiple potential interpretations for ambiguous rules, but only one unambiguous representation can be captured by the computer-readable form (Dimyadi *et al.*, 2016). Further research should better understand the ambiguity issue in building rules and propose suitable solutions.

5.5 Thorough understanding of rule texts is lacking

The classification methods in the literature can be grouped into the following types based on the main criteria for classification: semantics, complexity, interpretability with IFC, logic, syntactic, interdependency or a mixture of some of these. Although these classifications shed some light on building rules, there are still some deficiencies in these methods.

- Many classifications focus only on one aspect of building rules (e.g. semantics (Zhang *et al.*, 2022a)). However, as rules have different semantic constructs, logic features, complexity of checking and implicit assumptions, this could easily lead to a partial understanding of rules. Furthermore, when a comprehensive understanding of rules is lacking, it is easy to regard them as being only checkable manually. This limits the scope of using automated checking and ultimately affects checking efficiency.
- Many classifications are proposed solely to analyse whether a given rule can be checked automatically or not. Although this is an important goal of analysing rules, the classification could provide more benefits, such as identifying suitable representation methods based on the features of a specific type of rule.

Existing classification methods mainly focus on individual provisions. As a result, many have failed to recognise the importance of requirement titles when interpreting these rule provisions, regarding them as irrelevant or even ‘non-sense’ (Zhang and El-Gohary, 2021a). However, titles can provide an essential context or applicability for the rule provisions. In addition, while focusing on individual rule provisions, the linkages and interdependencies among sentences have mainly been ignored.

To address this issue, a consolidated list of aspects should be used to conduct the rule analysis to understand rules thoroughly. Specifically, the consolidated list should consider two levels. On the lower level, it concerns the constructs of individual rules. This may include the complexity of checking the rules, the level of ambiguity, the semantic constructs (e.g. conditions, requirements) and so on. The definition and scale of an individual rule also need to be made clear. On a higher level, the relationships among rule constructs and different rule clauses also need to be analysed.

5.6 Knowledge of relationships between representations and rule type is insufficient

The relationships between rules and their representations are reflected by the capabilities of the representation method and the capabilities needed to represent specific rules (Figure 5). Unfortunately, the current literature largely ignores the exploration of such relationships. As a result, the representation development process typically improved certain aspects of representation (e.g. conciseness) at the cost of others (e.g. expressiveness). This could be one of the reasons why there is still no representation capable

of representing all building rules, independent of the target technologies and schemas, such as BIM and/or geographic information systems (Altıntaş and Ilal, 2021). Therefore, a change is needed in the representation development process to address this issue. It is envisaged that the new process should ideally include four steps:

- (a) identify what rules are being represented
- (b) analyse and extract rule features
- (c) make comparisons of several potential representations, analysing their capabilities and what capabilities can be acquired with some valid adaptations
- (d) select and match rules and representations (make some adaptations if needed).

Using the proposed steps, developers will have a good idea of what capabilities they are looking for when they develop the representation method. The developed representation method can better fulfil the proposed goals by analysing the rule features and the relationships between rules and representations. The validity of the proposed steps needs to be verified by future research.

6. Conclusions

Automated code compliance checking is a promising method to accelerate the design review and approval process and help achieve better quality, productivity gain and cost savings in the AEC industry. However, one of the keys to ensuring a favourable ACC result is accurate and efficient rule capture. This paper adopted a systematic literature review method to review the literature on this topic; 124 papers were reviewed and analysed in depth. The results show that rule capture in ACC has gained popularity recently, and more semi-automatic and automatic methods have been proposed.

Despite intensive research in this domain, the rule capture process either is not fully automated or requires experts’ review. Reasons for that lie across the rules, representations and relationships between them. In this paper, the authors identified and explained six challenges to be addressed in future research, including (a) the lack of solid methodological backdrop in current rule representation development processes; (b) the inability to handle unknown values and to represent other side effects; (c) restriction of representations by the target data model and/or rule engine; (d) the lack of capability to deal with ambiguous rules; (e) incomplete understanding of rule texts; and (f) the lack of knowledge of relationships between representations and rule type. Potential solutions were also proposed. This research is significant to the research community and practitioners of the AEC industry by providing research directions and potential ideas for representation development in ACC.

This research has limitations. First, the literature is retrieved only from the Scopus database. More related literature may be found when other databases are included. Second, the keyword used for the literature search may not be complete and some work may be omitted. (a) Future work could include a more comprehensive

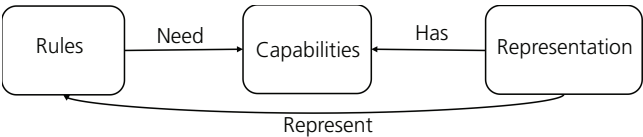


Figure 5. Relationships among rules, capabilities and representations

search by retrieving literature from Scopus, Web of Science and Google Scholar, and (b) more relevant keywords can be included for literature search.

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